

STAT 339 METHODS OF LINEAR ALGEBRA I

COURSE SYLLABUS

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(Department of Statistics & Actuarial Science)

October 10, 2023



UNIVERSITY OF GHANA

Outline

Outline

Lecturer Contact

- Office: Room 204.
- Email: eoacheampong@ug.edu.gh.
(Sending an email is the easiest way to contact me)
- Mondays: 08:30 – 12:30 GMT or by appointment.
- Phone number: +233 54 477 7777.
- Lecture hours: Monday 3:30 pm – 5:20 pm/ JQB09 & Thursday 6:30 pm – 7:20 pm/ LOT1

Outline

Teaching Assistant Contact

- Name: Mr Bless Forfoe
- Email: bforfoe@st.ug.edu.gh
- Office: Room 106
- Phone number: +233 24 977 8402.

Outline

Introduction/Subject or Course Overview

- The emphasis is on the geometric and computational foundations of Linear Algebra with abstraction (and proof) kept to a minimum. We would examine matrices, linear systems of equations and their solutions. Basic properties of determinants, vectors in R^n and a simple introduction to the idea of an arbitrary vector space. The pinnacle of the course is to be able to find eigenvalues and their corresponding eigenvectors for a given matrix and indeed for a linear map. We would carefully develop the diagonalization of symmetric matrices.

Outline

- This course is heavily reliant on your participation. The majority of mathematical topics and methods will be covered in class, and there will be lots of chances for questions and clarification. However, the finest lessons are those that come from practice and discussion. During outside of class, you must read the allocated text passages, complete the assigned exercises, and bring any questions to class or to the course tutorial section. The study of mathematics is not a spectator sport. Don't merely conduct computations; consider the challenges posed, your technique, the implications of your calculations, and the results you obtain. This will be extremely beneficial in class and on examinations..

- To provide students with instruction in this critical area of mathematics. Students are required to learn about matrices, linear equation systems, vector spaces, inner products, Eigen values, and linear transformations. Effort will be made to introduce the students to postulational and axiomatic approach in Mathematics. Additionally, the use of linear algebra in science and daily life is emphasized in this course..

Outline

Learning Outcomes

At the end of the course, the student should be able to:

- Use concept of elementary matrices to evaluate the inverse of square matrices and solving the system of linear equations
- Convert real life problems into system of linear equations and perform elementary row operations to solve them.
- Evaluate the determinant of matrix and solving system of equations using properties and/or operations on determinants.
- Analyse vectors and their properties in 2-space, 3-space and n-space.
- Recognize vector spaces and/or subspaces and compute their bases and its dimension.
- Perform Eigen Value analysis and use it to diagonalize a matrix and/or find its powers.
- Identify inner product spaces and/or perform Gram Schmidt process/QR decomposition using inner products.
- Express a linear transformations graphically and matrices and using their properties to solve statistical problems.

Outline

- Mode of course delivery will include lectures: videos, lecture notes, power point slides; tutorials and including presentations. There will be thirteen weeks of lectures for 3 hours a week

Outline

- Plagiarism in any form is unacceptable in the University of Ghana and shall be treated as a serious offence. Appropriate sanctions, as stipulated in the Plagiarism Policy of the University of Ghana Undergraduate Handbook, will be applied when students are found to have violated the Plagiarism policy.

Outline

- Popup Quizzes (5 %)
- Exercises (10 %)
- Mid-Semester Test/Interim Assessment (15 %)
- Final Exam (70 %)

NOTE

Grading Scale: Refer to the Undergraduate Handbook.

Outline

Reading List /Required Text

- Sheldon, A. (2010). Linear Algebra Done Right. Springer.
- Artin, M. (2010). Algebra (2nd Edition). Pearson Higher Ed USA.
- Roman, S.(2007). Advanced Linear Algebra. 3rd Edition. Springer.
- Strang, G. (2006). Introduction to Linear Algebra. Wellesley-Cambridge Press.
- Anton, H. & Rorres, C. (2013). Elementary linear algebra: applications version. John Wiley & Sons.
- Kolman, B and Hill, D.R. (2005). Introductory Linear Algebra, an applied first course, 8th edition, Addison Wesley.

Outline

- Disclaimer: This Course Outline may be subjected to changes as when necessary.
- It is expected of every student to be punctual (class attendance will be graded).
- Assignments are to be submitted by close of day (17:00 GMT) on or before the deadline which will be indicated on the assignment sheet.
- Students are required to comport themselves during lecture hours.
- Use of mobile phones during lecture hours is strictly prohibited.

Outline

Lecture Topics

- Matrices.
- Linear Equations in Linear Algebra.
- Determinants.
- Vector Spaces.
- Inner Product Spaces.
- Linear Transformations.
- Eigenvalues and Eigenvectors.

Outline

What is Algebra?

- Is the study of structure.
- Foundational branch of mathematics that you most likely covered at school.
- Very general term.

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 - Geometric
 - Applied topics: Actuarial Science, Engineering, Computer Science, Statistics, etc

Outline

- Very useful in modern society – Scientific research, engineering...
- Fundamental course for many other courses
 - other mathematical courses,
 - courses in science, technology, finance and management.
- Train you to have good abstractly reasoning ability. To think abstractly and mathematically.

- Makes complex computations easier
- See it in many different examinations
 - Financial Engineering,
 - Survival Models,
 - Enterprise Risk Management,
 - ...